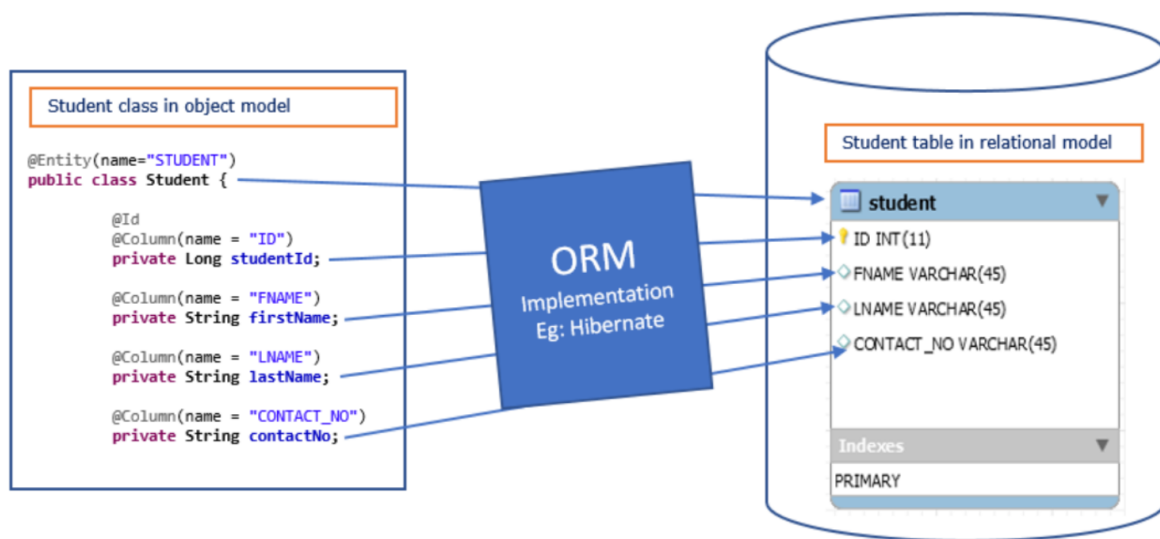
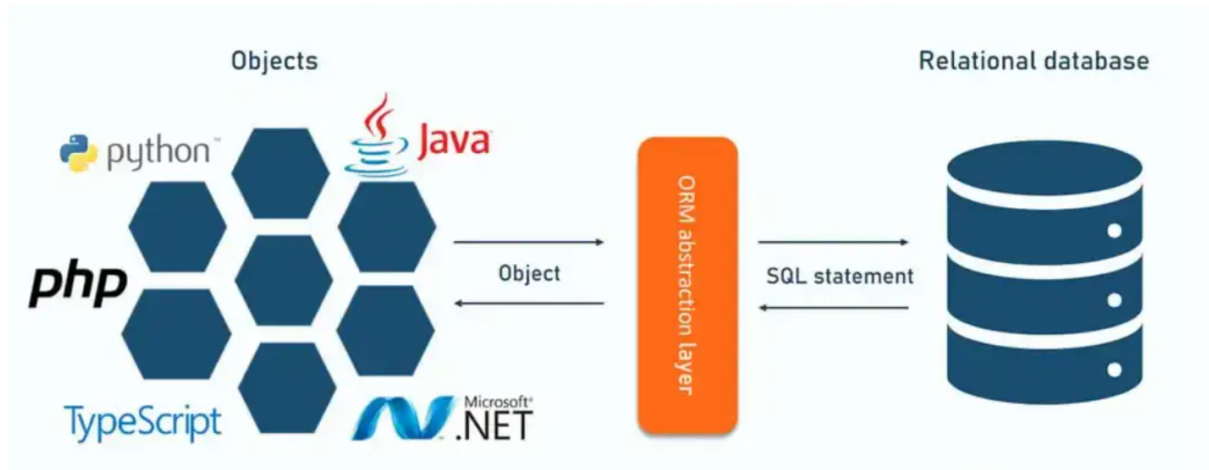


ORM



Class <--> Table

Fields <----> columns of table

ORM generates DDL and DML based on the mapping

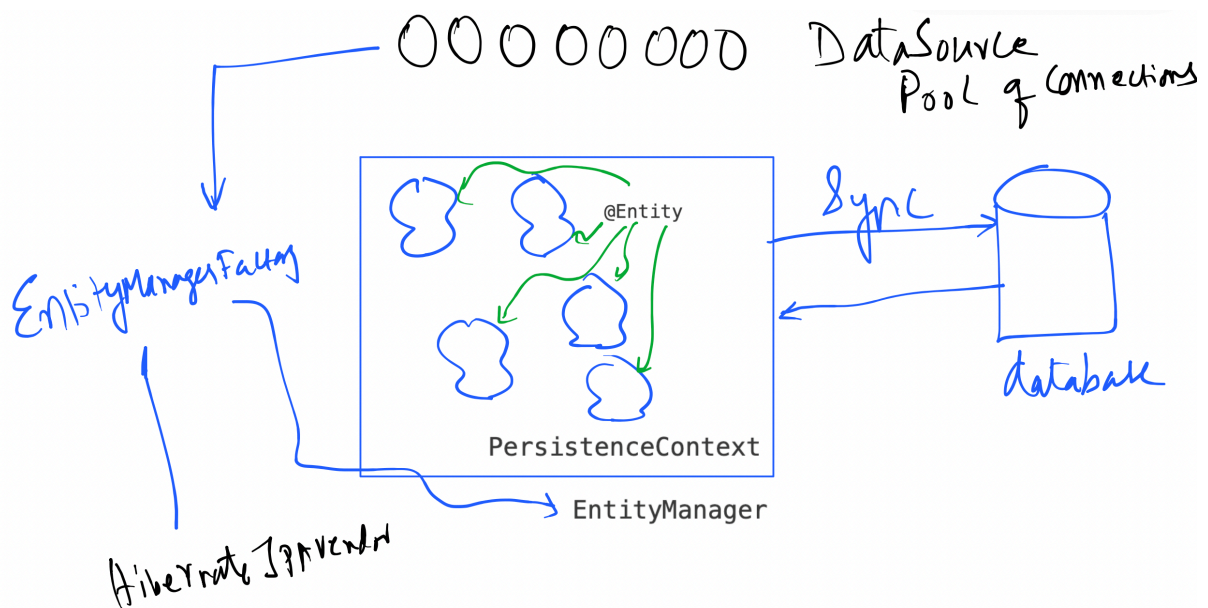
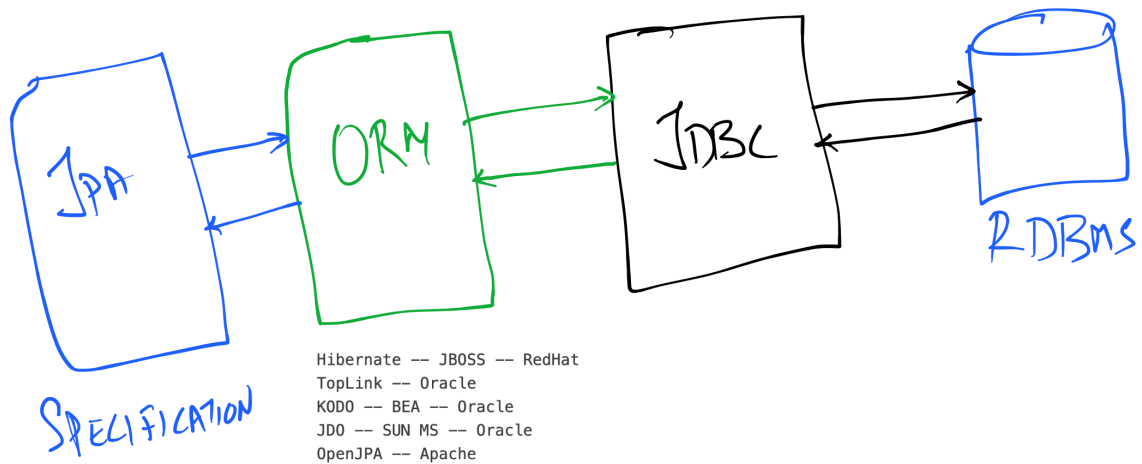
ORM vs JDBC

JDBC

```
@Override
public void addProduct(Product product) throws PersistenceException {
    final String SQL = "INSERT INTO products (id, name, price) VALUES (0, ?, ?)";
    Connection con = null;
    try {
        con = DBUtil.getConnection();
        PreparedStatement ps = con.prepareStatement(SQL);
        ps.setString(1, product.getName());
        ps.setDouble(2, product.getPrice());
        ps.executeUpdate();
    } catch (SQLException e) {
        if (e.getErrorCode() == 1062) {
            throw new PersistenceException("unable to add product, product with id "
                + product.getId() + " already exists", e);
        }
        throw new PersistenceException("unable to add product", e);
    } finally {
        DBUtil.closeConnection(con);
    }
}
```

ORM

```
@Override
public void addProduct(Product product) {
    em.persist(product);
}
```



```
[mysql> select * from customers;
```

email	fname	lname
anne@cisco.com	Anne	Hathaway
danny@cisco.com	Danny	Peter
ria@cisco.com	Ria	Patel

```
select * from customers
```

```
select * from customers where lname='Patel'
```

```
select email, fname from customers;
```

SQL uses table and column names

```
@Entity
@Table(name="customers")
public class Customer {
    @Id
    private String email;

    @Column(name="FNAME", length = 100)
    private String firstName;

    @Column(name="LNAME", length = 100)
    private String lastName;
}
```

```
from Customer
```

```
from Customer where lastName = 'Patel'
```

```
select email, firstName from Customer
```

JP-Ql uses class and field names; case sensitive; Polymorphic